

# ShopZada Data Warehouse Project

## Building an Enterprise-Grade Analytical Platform

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### **Team Members:**

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## Critical Issues

- Manual Excel analysis (hours of work)
- No historical trend analysis
- Inconsistent data quality
- Cannot answer cross-functional questions
- Delayed business decisions

## Fragmented Data

- 20+ files in 7 different formats (CSV, JSON Excel, HTML, Parquet, Pickle)
- Data scattered across 5 departments (Customer, Operations, Marketing, Enterprise, Business)
- 1.5M+ orders with no unified view

## Business Questions to Answer

- What are our top products by revenue?
- Which customers contribute most?
- What campaigns drive the highest ROI?
- How do merchants perform?

# ShopZada's Data Challenges

# The Solution: Unified Data Warehouse

## BEFORE

- 20+ disparate files
- Hours to answer questions
- Manual data compilation
- No single source of truth

## AFTER

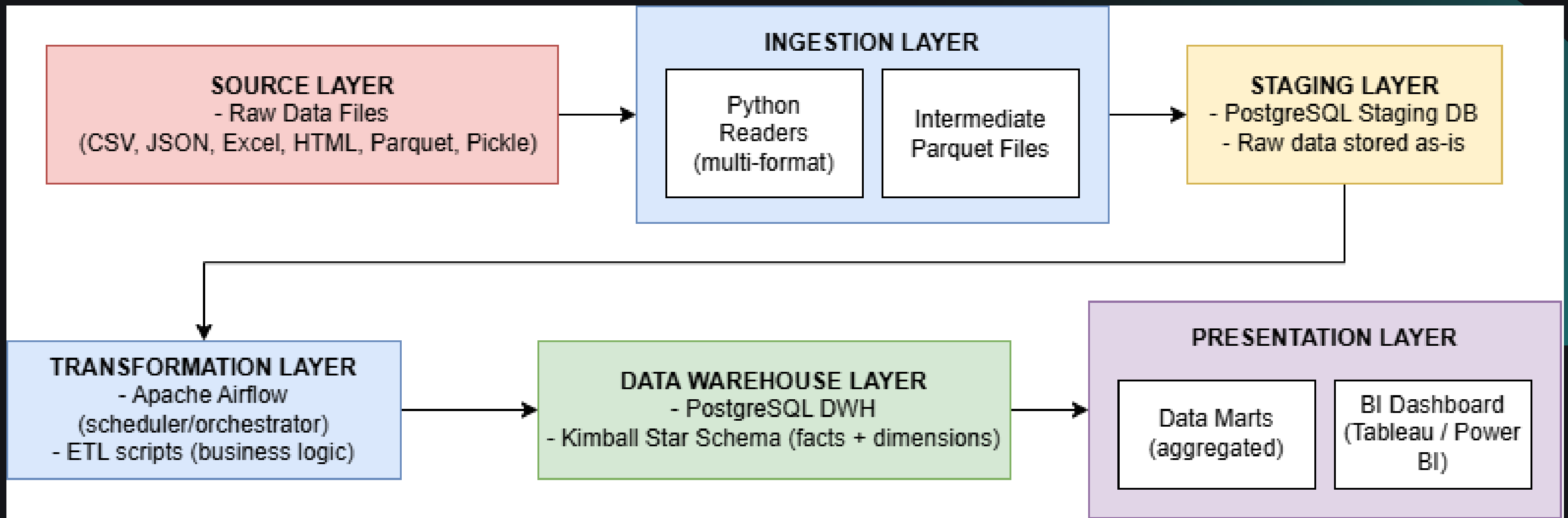
- Single consolidated database
- Seconds to generate insights
- Automated ETL pipeline
- 4+ years of historical data (2020-2024)

## METHODOLOGY

- Kimball Dimensional Modeling
- Star schema design for optimal analytics
- Business-focused approach
- Proven industry standard



# System Architecture: End-to-End Data Platform



# Technical Infrastructure Fully Containerized with Docker

Airflow (Web UI, Scheduler,  
Worker, DAG Processor)

Staging  
DB

DWH  
DB

Redis  
Queue

One-Command Deployment:  
`docker compose up -d`

## Benefits:

- Reproducible environment
- Easy deployment
- Scalable infrastructure
- Isolated services



# Tech Stack

| Component        | Technology     | Justification                                            |
|------------------|----------------|----------------------------------------------------------|
| Orchestration    | Apache Airflow | Industry standard for workflow automation                |
| Data Warehouse   | PostgreSQL     | Open-source RDBMS with excellent analytical capabilities |
| Staging Database | PostgreSQL     | Same tech for consistency                                |
| Message Queue    | Redis          | Celery backend for distributed task execution            |
| Containerization | Docker         | Reproducible deployment                                  |
| ETL Language     | Python         | Rich ecosystem for data engineering                      |
| BI Tool          | Metabase       | Open-source dashboard tool that allows docker setup      |



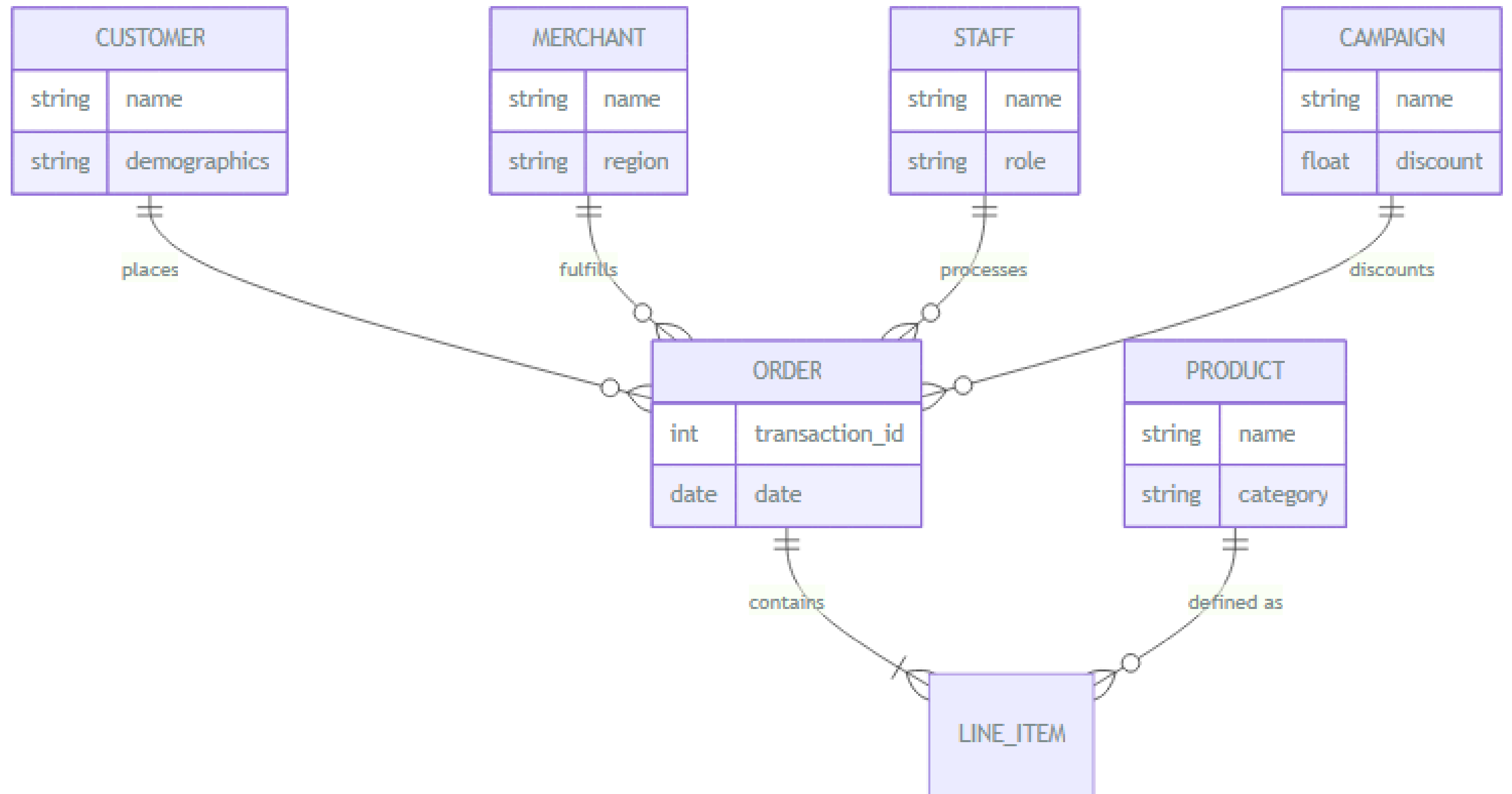
# Data Inventory: What Data Do We Have?

| Domain    | Records | Time Range   | Formats   |
|-----------|---------|--------------|-----------|
| Orders    | 1.5M+   | 2020-2024    | 5 formats |
| Customers | 50K+    | 2021-present | 3 formats |
| Products  | 20K+    | current      | Excel     |
| Merchants | 10K+    | 2020-present | 2 formats |
| Staff     | 30K+    | 2020-present | HTML      |
| Campaigns | 50K+    | 2023-present | CSV       |

- Data Quality: All Successfully Staged**
- 100% of source files ingested
  - Data validation checks passed
  - Ready for transformation

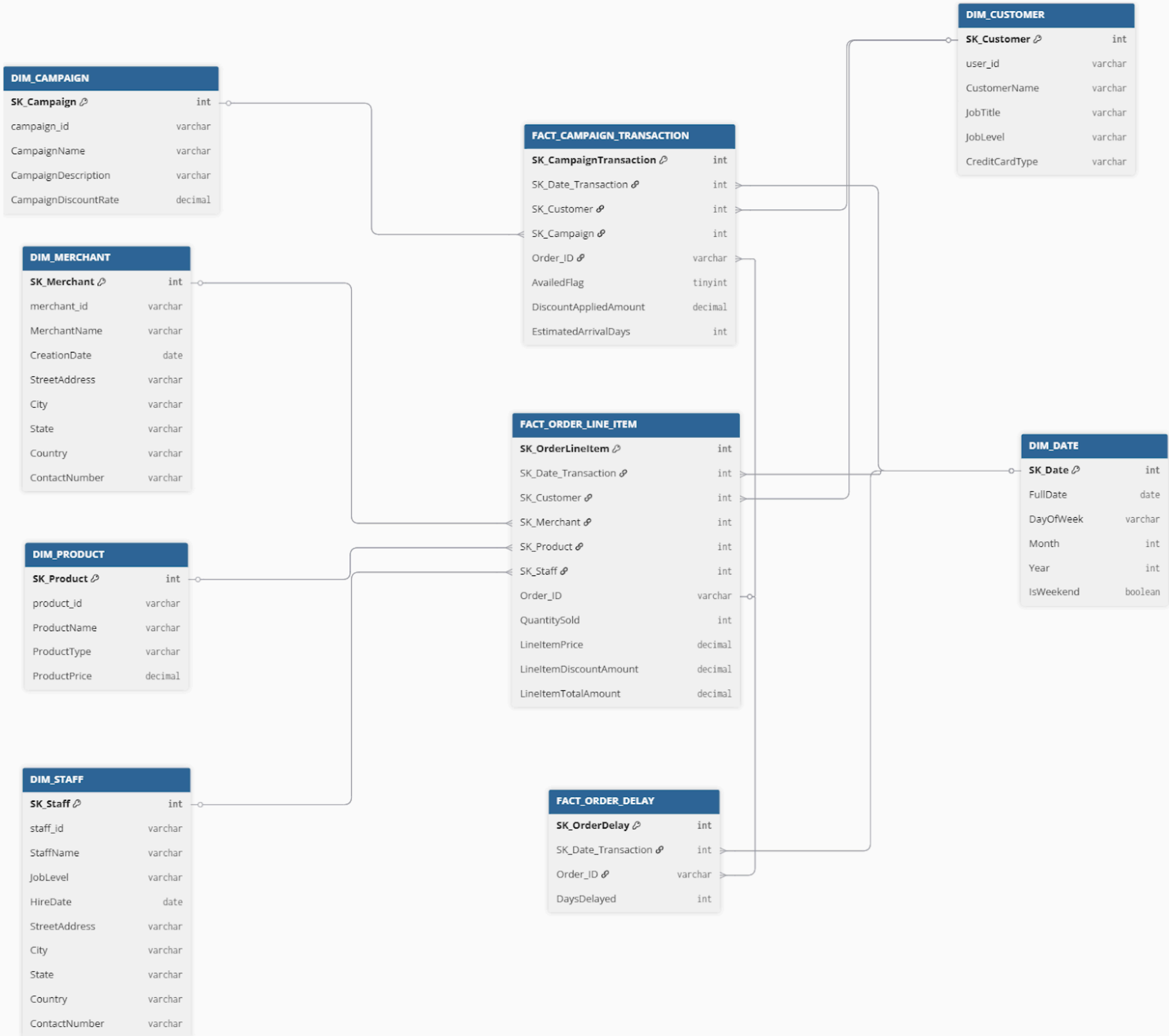
# Conceptual Data Model

High-level relationships between ShopZada business entities.





# Logical Data Model



# Physical Data Model

fact\_order\_line\_item

| Column               | Type    | Key | Description       |
|----------------------|---------|-----|-------------------|
| sk_orderlineitem     | INT     | PK  | Surrogate key     |
| order_id             | TEXT    | NK  | Business order ID |
| sk_customer          | INT     | FK  | Customer          |
| sk_merchant          | INT     | FK  | Merchant          |
| sk_staff             | INT     | FK  | Fulfillment staff |
| sk_product           | INT     | FK  | Product           |
| sk_date              | INT     | FK  | Transaction date  |
| quantitysold         | INT     | —   | Units ordered     |
| lineitemprice        | DECIMAL | —   | Unit price        |
| lineitemtotal amount | DECIMAL | —   | qty × price       |

fact\_campaign\_transaction

| Column               | Type    | Key | Description       |
|----------------------|---------|-----|-------------------|
| sk_campaigntrans     | INT     | PK  | Surrogate key     |
| order_id             | TEXT    | NK  | Order ID          |
| sk_campaign          | INT     | FK  | Campaign          |
| sk_date              | INT     | FK  | Transaction date  |
| availedflag          | BOOLEAN | —   | Campaign used     |
| estimatedarrivaldays | INT     | —   | Delivery estimate |

fact\_order\_delay

| Column        | Type | Key | Description    |
|---------------|------|-----|----------------|
| sk_orderdelay | INT  | PK  | Surrogate key  |
| order_id      | TEXT | NK  | Order ID       |
| daysdelayed   | INT  | —   | Delay duration |
| sk_date       | INT  | FK  | Order date     |

Physical  
Data Model

dim\_customer

| Column         | Type | Key | Description    |
|----------------|------|-----|----------------|
| sk_customer    | INT  | PK  | Surrogate key  |
| customer_id    | TEXT | NK  | Source user ID |
| customername   | TEXT | —   | Full name      |
| jobtitle       | TEXT | —   | Occupation     |
| joblevel       | TEXT | —   | Seniority      |
| creditcardtype | TEXT | —   | Payment type   |

dim\_product

| Column       | Type    | Key | Description   |
|--------------|---------|-----|---------------|
| sk_product   | INT     | PK  | Surrogate key |
| product_id   | TEXT    | NK  | Product ID    |
| productname  | TEXT    | —   | Name          |
| producttype  | TEXT    | —   | Category      |
| productprice | DECIMAL | —   | Base price    |

Design Notes: Star schema implemented using Kimball methodology. Fact tables store quantitative measures; dimensions provide descriptive context using surrogate keys for performance.

# Physical Data Model

dim\_merchant

| Column       | Type | Key | Description   |
|--------------|------|-----|---------------|
| sk_merchant  | INT  | PK  | Surrogate key |
| merchant_id  | TEXT | NK  | Merchant ID   |
| merchantname | TEXT | —   | Business name |
| city         | TEXT | —   | Location      |
| country      | TEXT | —   | Country       |

dim\_campaign

| Column        | Type | Key | Description   |
|---------------|------|-----|---------------|
| sk_campaign   | INT  | PK  | Surrogate key |
| campaign_id   | TEXT | NK  | Campaign ID   |
| campaignname  | TEXT | —   | Name          |
| discount_rate | TEXT | —   | Discount      |

dim\_date

| Column      | Type | Key | Description |
|-------------|------|-----|-------------|
| sk_date     | INT  | PK  | YYYYMMDD    |
| fulldate    | DATE | —   | Full date   |
| monthnumber | INT  | —   | Month       |
| quarter     | INT  | —   | Quarter     |
| year        | INT  | —   | Year        |
| dayname     | TEXT | —   | Day of week |

## Dashboards & Analytics

### Dashboard 1: Executive Overview

- **KPIs:** Revenue, Orders, Customers
- **Charts:** Sales trends, top products, geographic distribution

### Dashboard 2: Sales Performance

- Top products by revenue
- Product category analysis
- Year-over-year comparisons

### Dashboard 3: Customer Analytics

- Customer demographics
- Purchase patterns
- Customer lifetime value

### Dashboard 4: Marketing ROI

- Campaign performance comparison
- Discount effectiveness
- Conversion metrics



## Sample Analytics: What Business Questions Can We Answer?

### Example Query 1: Top Products

```
SELECT
  product_name,
  SUM(quantity * unit_price) as
revenue,
  SUM(quantity) as units_sold
FROM fact_orders f
JOIN dim_product p ON
f.product_key = p.product_key
WHERE year = 2024
GROUP BY product_name
ORDER BY revenue DESC
LIMIT 10;
```

### Example Query 2: Campaign ROI

```
SELECT
  campaign_name,
  COUNT(DISTINCT order_id)
as orders,
  SUM(discount_amount) as
cost,
  SUM(revenue) as
revenue_generated
FROM
fact_campaign_performance
GROUP BY campaign_name;
```



# Project Summary

## What we delivered

- Enterprise data warehouse infrastructure
- Automated ETL pipeline (Airflow orchestration)
- 1.5M+ transactions staged and validated
- Kimball dimensional model design
- Scalable, containerized platform
- Comprehensive documentation

## Business Impact

- Single source of truth for data-driven decisions
- Historical analysis capability
- Fast insights via automated dashboards
- Scalable platform ready for growth